

Data Sheet

Customer : _____

Product : Aluminum Electrolytic Capacitors – EXR Series _____

Size : 5x11mm ~ 22x40mm _____

Issued Date : 07-Apr.-2023 _____

Edition : Ver. 3 _____

Record of change

Date	Ver.	Description	Page
25-May-2016	1		
23-Apr.-2018	2	Load life 100V:2000hrs	1
07-Apr.-2023	3	Add 160V~450V Impedance	3

HITANO ENTERPRISE CORP.

7F-7, No. 3, Wu Chuan 1st Road, New Taipei Industrial Park,
New Taipei City, TAIWAN, R.O.C.

Tel: +886 2 2299 1331 (Rep.)

Fax: +886 2 2298 2466, 2298 2969

Prepared by	Checked by	Approved by	Accepted by (customer)
07-Apr.-2023	07-Apr.-2023	07-Apr.-2023	
<i>Randy Yu</i>	<i>Michelle Lin</i>	<i>Arthur Su</i>	

Subject : Storage of Aluminium Electrolytic Capacitors

We recommend the following conditions for storage :

1. It is recommended to keep capacitors between the ambient temperatures of 5°C to 35°C and a relative humidity of 75% or below.
2. Confirm that the environment does not have any of the following conditions :
 - (1) Damp conditions such as water, saltwater spray, or oil spray or fumes. High humidity or humidity condensation situations.
 - (2) In an atmosphere filled with toxic gasses (such as hydrogen sulfide, sulfurous acid, nitrous acid, chlorine, ammonia, etc.)
 - (3) Being exposed to direct sunlight, ozone, ultraviolet ray, or radiation.
 - (4) Being exposed to acidic or alkaline solutions.
3. Keep capacitors in the original package.

4. Storage life & Re-aging :

When Aluminium Electrolytic Capacitors are stored without applied voltage, their L.C.

(Leakage Current) characteristic increases over time. For long-term stored products, the following treatments must be performed before use :

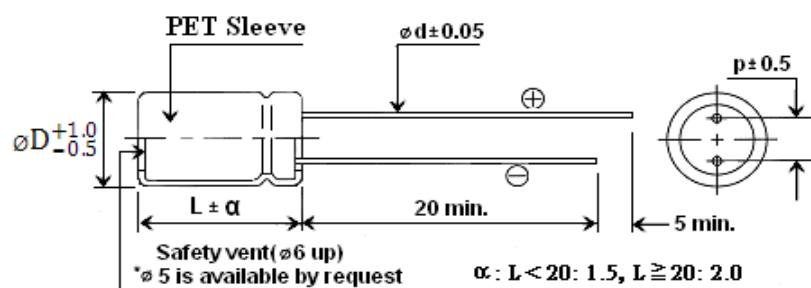
- (1) For Low Voltage Aluminium Electrolytic Capacitors (i.e., Working Voltage W.V. $\leq$ 120V) :

After one year of storage, a test must be performed before use. If the L.C. value exceeds the specified value, it is recommended not to use them, as lifespan and quality cannot be 100% guaranteed.
- (2) For Medium/High Voltage Aluminium Electrolytic Capacitors (i.e., Working Voltage W.V. $\geq$ 160V) :
 - (A) If stored for more than 6 months, a test must be performed before use to ensure lifespan and quality.
 - (B) If stored for 6-24 months and the L.C. value is between 25% and 40% of the specified value, it is recommended to recharge (re-agent) before use. If the L.C. value exceeds 40% of the specified value, do not use.
- (3) Re-aging condition : It is recommended to apply D.C. working voltage to the capacitor for 2 hours through 1K Ω of protective series resistor.

- EXR series capacitors are ideal for use in switching power supplies, communication equipments and etc.
- **Low Impedance** and long life.
- Safety vent construction design.
- RoHS Compliant

Characteristics

Voltage Range	6.3 to 100 VDC					160 to 450 VDC			
Capacitance Range	4.7 to 15000uF					1 to 470uF			
Temperature Range	-40 to +105°C					-25 to +105°C			
Leakage Current	I ≤ 0.01CV or 2uA, whichever is greater 3 minutes after Rated Voltage applied					I ≤ 0.03CV 3 minutes after Rated Voltage applied			
Capacitance Tolerance	±20% at 120Hz , 20°C (10% Tol. is available upon request)								
Dissipation Factor (at 20°C, 120Hz)	Rated Voltage (V)	6.3	10	16	25	35	50	63	100
	Dissipation Factor (tanδ) max	0.18	0.16	0.14	0.12	0.10	0.09	0.08	0.08
	Rated Voltage (V)	160	200	250	350	400	450		
	Dissipation Factor (tanδ) max	0.12	0.12	0.12	0.15	0.15	0.17		
Low Temperature Characteristics (120Hz)	For capacitance > 1000uF, add 0.02 for every 1000uF								
	Working Voltage (V)	6.3	10	16	25	35	50	63	100
	Z-25°C/Z +20°C	4	3	3	3	3	3	2	2
	Z-40°C/Z +20°C	8	6	4	3	3	3	3	3
	Working Voltage (V)	160	200	250	350	400	450		
	Z-25°C/Z +20°C	2	2	3	5	5	6		
	Z-40°C/Z +20°C	3	6	6	6	6	-		
For capacitance > 1000uF, add 0.5 every 1000uF for -25°C/+20°C add 1.0 every 1000uF for -40°C/+20°C									
Load Life		After the rated voltage with ripple current has been applied for at 105°C (100V~450V : 2000HRS)	Capacitance change			Within ±20% of initial value			
D §	Life Hours		D.F. tanδ			200% or less of initial specified value			
5 – 6.3 §	2000								
8 §	3000								
≥ 10 §	5000				Leakage current			Less than initial specified value	
Shelf life (at 105°C)		After storage for 1000 hours at 105°C with no voltage applied, the capacitor shall meet the specified limit in load life. Pre-treatment for measurement shall be conducted after application of DC working voltage for 30 minutes.							



Drawing

Dφ	5	6.3	8	10	13	16	18
p	2.0	2.5	3.5	5.0	5.0	7.5	7.5
dφ	0.5	0.5	0.5	0.6	0.6	0.8	0.8

Ripple Current Coefficients

Frequency (Hz)	50(60)	120	400	1K	10K	100K
Cap.(uF) / Hz	Multiplier					
Cap. ≤ 10	0.47	0.59	0.76	0.85	0.97	1
10 < Cap. ≤ 100	0.52	0.62	0.80	0.89	0.97	1
100 < Cap. ≤ 1000	0.58	0.72	0.84	0.90	0.98	1
1000 < Cap.	0.63	0.78	0.87	0.91	0.98	1

Case size & Maximum Ripple Current (mA rms 105°C, 100KHz) & Imp. (Ω 20°C, 100KHz)

WV Cap	6.3			10			16			25		
uF	Size	Imp	RC	Size	Imp	RC	Size	Imp	RC	Size	Imp	RC
10							5x11	4.70	74	5x11	2.56	85
22							5x11	2.60	100	5x11	1.95	115
33							5x11	2.00	114	5x11	1.42	155
47							5x11	1.10	155	5x11	1.10	155
68							5x11	0.69	195	6.3x11	0.65	230
100				5x11	0.80	210	6.3x11	0.50	265	6.3x11	0.35	370
150				6.3x11	0.61	290	8x12	0.41	370	8x12	0.31	390
220	6.3x11	0.65	310	6.3X11	0.35	340	8x12	0.25	450	8x12	0.15	555
330	8x12	0.42	390	8x12	0.27	460	8x12	0.15	590	8x14	0.12	700
470	8x12	0.25	450	8x12	0.25	580	10x12.5	0.12	750	10x16	0.076	1010
560							10x12.5	0.1	800	10x12.5	0.07	1105
680	8x14	0.21	550	8x14	0.11	765	10x16	0.080	850	10x21	0.065	1220
820	8x12.5	0.2	620	10x16	0.1	890	10x21	0.078	1180	10x21 13x26	0.052 0.055	1400 1960
1000	8x12 10x12.5	0.17 0.10	555 750	10x12.5 10x21	0.080 0.076	805 1040	10x21	0.068	1200	13x21	0.045	1650
1200	10x16	0.16	860	10x16	0.067	1200	10x21	0.061	1500	13x21	0.041	1980
1500	8x16 10x16	0.14 0.08	830 940	10x16 10x21	0.070 0.062	1000 1260	13x21	0.045	1575	13x26	0.038	1960
1800	10x16	0.11	1250	10x21	0.058	1550	13x21	0.047	1800	13x25	0.038	2210
2200	10x21	0.068	1050	10x21	0.052	1260	13x21	0.039	1900	16x26 13x26	0.036 0.044	2100 1860
3300	10x21 13x21	0.052 0.060	1260 1650	13x21	0.039	1660	16x26	0.030	1950	16x31.5	0.026	2680
3900				13x26	0.030	1720	16x26	0.030	2180			
4700	13x21	0.039	1900	13x26	0.030	1950	16x31.5	0.022	2410	18x36	0.018	2960
6800	16x26	0.032	2350	16x26	0.022	2150	16x36	0.018	2400	18x41	0.015	3550
10000	16x31.5	0.024	2700	16x31.5	0.018	2400	18x36	0.015	2800	22x41		
15000	18x36	0.023	2950									

Case size & Maximum Ripple Current (mA rms 105°C 100KHz) & Imp. (Ω 20°C 100KHz)

WV Cap.	35			50			63			100		
uF	Size	Imp	RC	Size	Imp	RC	Size	Imp	RC	Size	Imp	RC
1				5x11	3.5	40	5x11	3.0	20	5x11	4.4	30
2.2				5x11	3.5	40	5x11	2.8	35	5x11	3.3	42
3.3				5x11	3.5	40	5x11	2.2	50	5x11	2.60	72
4.7				5x11	2.00	90	5x11	2.00	65	5x11	2.60	72
6.8				5x11	1.89	110	5x11	1.82	100	6.3x11	1.77	130
10	5x11	2.2	107	5x11	1.82	120	5x11	1.75	110	6.3x11	1.77	130
22	5x11	1.5	150	6.3x11	1.25	150	6.3x11	0.80	240	8x12	0.85	220
33	5x11	1.21	180	6.3x11	0.80	250	6.3x11	0.61	270	10x12.5 10x16	0.70 0.69	293 320
47	6.3x11	0.80	250	6.3x11	0.65	290	8x12	0.56	300	10x12.5 10x16	0.58 0.37	370 382
68	6.3x11	0.64	280	8x12	0.33	375	10x12.5	0.21	480	10x16 10x21	0.35 0.28	470 501
100	8x12	0.25	450	10x12.5	0.17	480	10x16 10x12.5	0.14 0.17	530 535	13x21	0.18	714
120										13x16 13x21	1.08 1.14	90 90
150	8x12	0.191	510	10x12.5	0.132	560	10x16	0.11	600	13x21 16x16	0.174 0.15	780 820
220	10x16 10x12.5	0.114 0.17	750 620	10x16	0.096	630	10x21	0.08	710	13x26 16x26	0.13 0.10	950 1282
270	10x12.5	0.17	620				13x21	0.055	1250	13x30	0.11	1120
330	10x16	0.079	1050	10x21	0.078	960	13x21 13x26	0.055 0.055	1250 1350	16x26 16x31.5	0.10 0.09	1440 1563
470	10x21	0.065	1200	13x21	0.055	1400	13x26	0.053	1620	16x31.5 18x32	0.09 0.076	1650 1907
560				13x21	0.049	1640	13x26	0.049	1680	18x36	0.068	2000
680	13x21	0.056	1570	13x26	0.044	1830	16x26 13x30	0.043 0.045	1950 2160	18x36 16x41	0.08 0.083	1790 1790
820	13x21	0.048	1700	13x30	0.039	2100	16x26	0.038	2150	18x36	0.071	1840
1000	13x26	0.042	1900	16x26	0.036	2300	16x31.5	0.034	2350	18x41	0.066	1930
1200	10x21	0.039	2130	13x26			16x31.5	0.04	2560			
1500	16x26	0.026	2490	16x31.5	0.034	2650	18x36	0.031	2710	22x41	0.022	3720
1800	16x26	0.024	2520				18x41	0.03	3000			
2200	16x31.5	0.022	2550	18x36	0.032	3070	18x41	0.024	3600	25x50	0.046	4680
2700	13x35	0.02	2610									
3300	16x36	0.016	2650	18x41	0.025	3100	22x46	0.024	3950			
4700	18x41	0.010	3000	22x40	0.022	3720						

Case size & Maximum Ripple Current (mA rms 105°C 100KHz) & Imp. (Ω 20°C 100KHz)

WV Cap	160			200			250			350		
uF	Size	Imp	RC	Size	Imp	RC	Size	Imp	RC	Size	Imp	RC
1	6.3x11	7.85	45	6.3x11	7.76	45	6.3x11	6.54	50	8x12	6.35	58
2.2	6.3x11	5.21	55	6.3x11	5.18	55	8x12 6.3x11	4.12	72	8x12 10x12.5	5.3 4.02	75 86
3.3	8x12	4.31	70	8x12	4.25	71	8x12	3.85	75	10x12.5	3.8	90
4.7	8x12	4.16	72	8x12 10x12.5	5.0 4.12	78 85	8x12 10x12.5	3.50 2.95	85 100	10x21 10x12.5	6.70	130
10	10x12.5 10x16	3.00 2.69	126 140	8x12 10x16	3.75 2.95	115 132	10x12.5 10x16	3.10	160	10x21 13x21	4.65	200
22	10x16 10x21	1.30 2.10	185 205	10x16 10x21	1.80 1.51	186 205	10x16 10x21	1.52	185	13x21	2.60	220
33	13x21	1.30	260	10x21 13x21	1.30 0.80	280 330	13x21	1.45	310	13x26	1.78	290
47	10x21 13x21	1.65 1.38	276 320	13x21 13x26	1.275 1.10	360 400	13x21 13x26 10x26	1.20	405	16x26 16x31.5	1.51	430
68	13x26	0.62	450	13x26 16x26 16x21	0.60	540	13x26 16x26	0.38	490	16x31.5	1.10	475
100	16x26	0.47	540	16x26 16x31.5	0.435	820	16x26 16x31.5	0.315	675	18x36	0.70	513
150	16x31.5	0.43	710	16x36	0.23	860	16x36	0.24	750	18x45	0.50	590
220	16x36	0.256	820	18x36 18x41	0.525 0.19	1050 1090	18x41	0.28	910			
330	18x41	0.195	1180									

Cap	wv	400			450		
	uF	Size	Imp	RC	Size	Imp	RC
	1	8x12	16.5	36	8x12	17.35	45
	2.2	8x12 10x12.5	13.0 13.0	65 76	10x12.5 10x16	10.25	65
	3.3	8x12 10x12.5	12.0 21.2	86 105	10x12.5 10x16	18.2	89
	4.7	10x12.5 10x21	11.0 10.0	105 120	10x16 13x21	6.85	110
	10	10x21 13x21	6.30 5.45	218 235	13x21 13x26	5.60 6.30	180 189
	22	13x21 13x26	3.15 2.65	268 295	13x26	2.8	320
	33	13x26 16x26	1.60 2.90	399 440	16x26	2.2	460
	47	16x26 16x31.5	1.60 1.56	539 580	16x36	1.35	650
	68	18x25 18x32	0.99 1.10	774 800	18x36	1.08	760
	100	18x36 18x41	0.70 0.72	854 900	18x36 18x41	1.10 1.05	825 880
	150	22x40	0.44	1180	22x40	0.48	998
	180				25x40	0.4	1120
	220	25x40	0.4	1340			

Part Numbering System

EXR	101	M	25	A	—	T1
SERIES	CAPACITANCE	TOL.	W.V.	PACKAGE	SIZE	LEAD SPACE
	IN 3DIGITS	M= ± 20%	0J= 6.3V	B= Bulk	Omit if only	Omit if Bulk
	010= 1.0uF		10= 10V	C5= Cut 5mm	one size	T1= L/S 2.5mm Taped
	4R7= 4.7 uF		16= 16V	AC5= Smaller Size cut 5mm	A= Smaller Size	TA= Lead forming space 5mm Taped
	101= 100uF		25= 25V			
	102= 1000uF		35= 35V	A= Ammo Pack		T35= L/S 3.5mm Taped
			50= 50V	R= Tape&Reel		T2=L/S 5mm Taped
			63= 63V	F5= Lead formed & cut 5mm		T3= L/S 7.5mm Taped
			2A= 100V			
			2C= 160V			
			2D= 200V			
			2E= 250V			
			2V= 350V			
			2G= 400V			
			2W= 450V			